CCSA101 LAB01 REPORT TEMPLATE

ICDFA / Check Point SecureAcademy

CCSA101-LAB01: Check Point Lab Setup and Administration

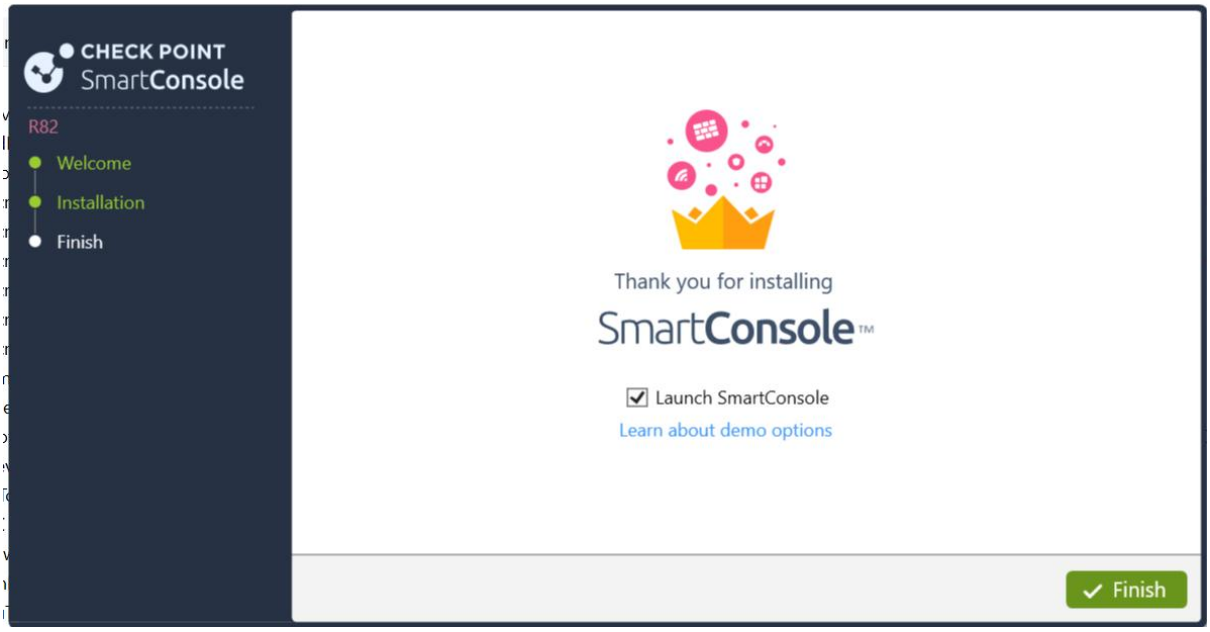
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Course: CCSA101 Introduction to Quantum Security
Submission Date: 30/10/2026

Part A: Prepare VMware and the private network

Step 1: Install the Windows tools

Objective

install VMware Workstation Pro and R82 SmartConsole.



A1

Item: R82 SmartConsole
Version: 6.11.0
Source : checkpoint

Item: VMware Workstation Pro
Version: 26.0.0
Source: Getintopc

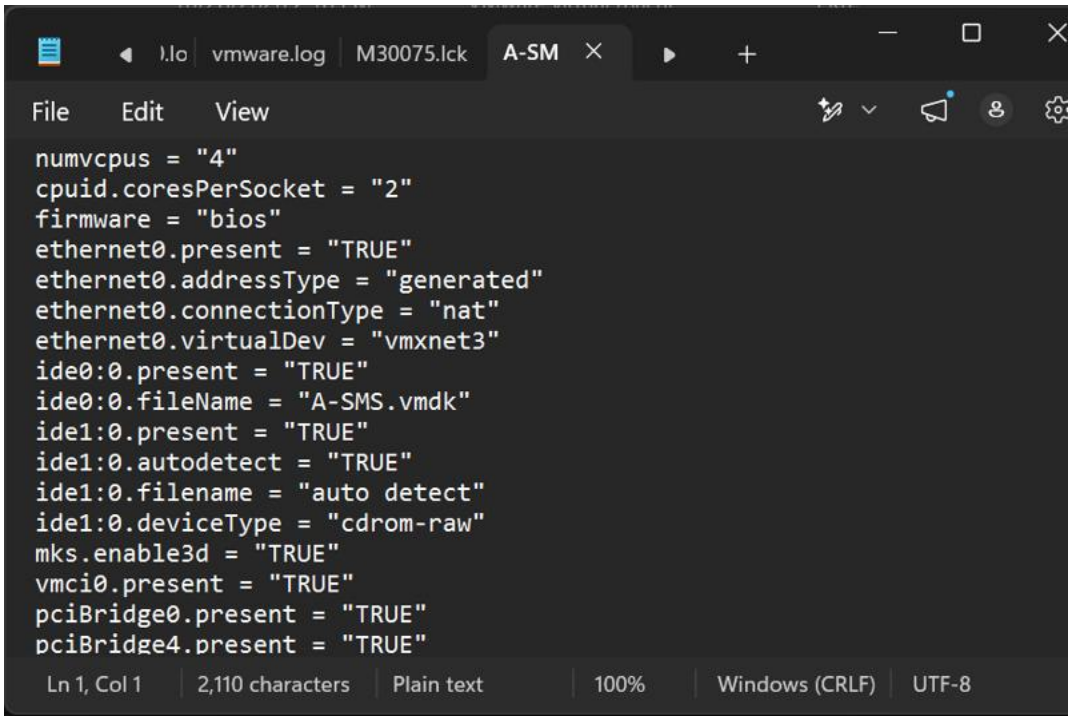
Interpretation

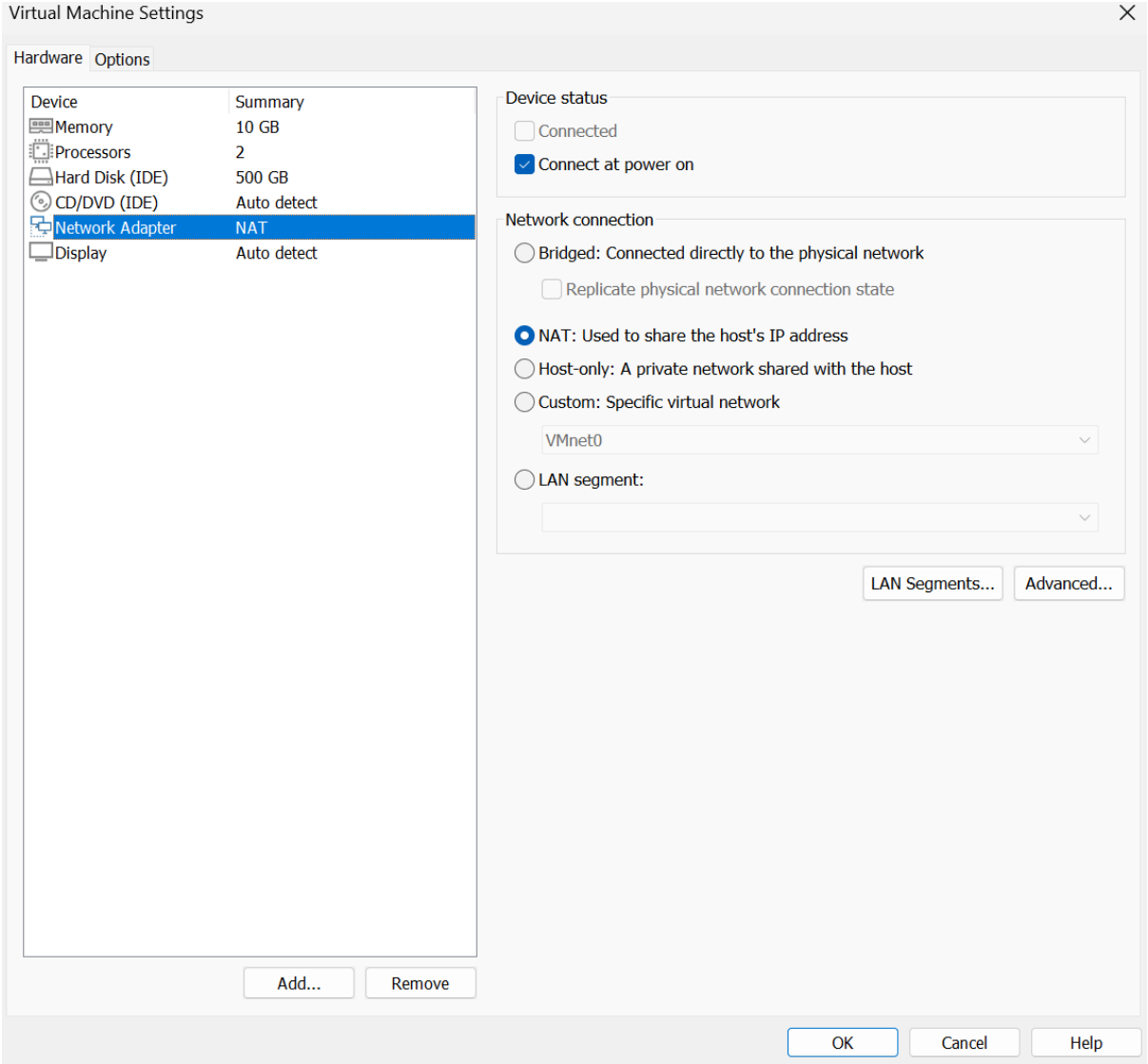
Both required software packages were installed successfully

Step 2: Load A SMS and check its adapter

Objective

Verify A-SMS configuration and VMXNET3 adapter





A2

Interpretation

The VM is configured according to lab requirement

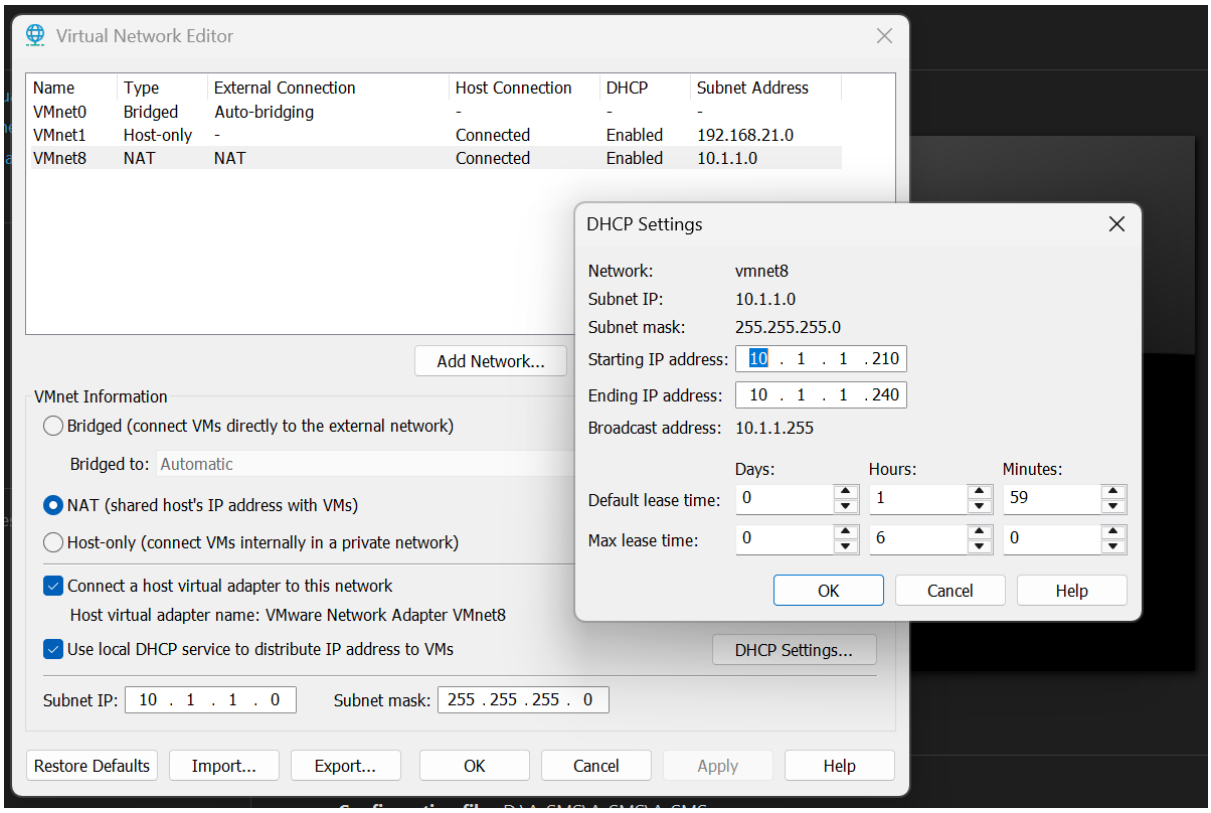
Step 3: Set the VMnet8 network

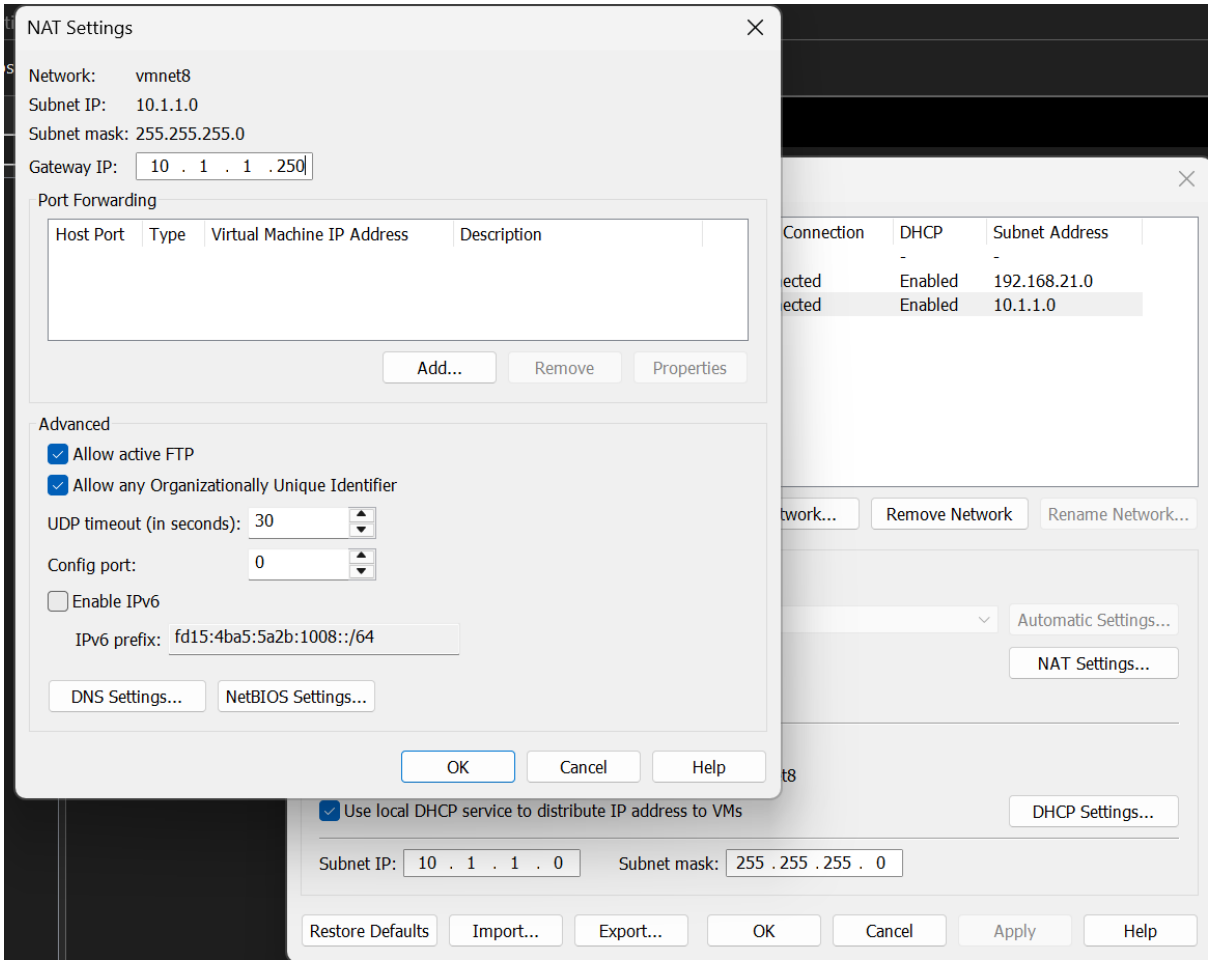
Required settings

Subnet: 10.1.1.0/24

NAT Gateway: 10.1.1.250

DHCP Pool: 10.1.1.210 – 10.1.1.240





A3

Interpretation

VMnet8 provides a private management network independent of the physical Wi-Fi connection.

Part A Set the addresses and verify Gaia

Step 4: Set the Windows VMnet8 address

Required settings

VMnet8 = 10.1.1.201/24

Observation

IPv4 Address. . . . . : 10.1.1.201



A4

Step 5: Boot A SMS and inspect eth0

Command used

show interface eth0

Observation

Ip address: 10.1.1.101

Sate: ON

Link Status: UP

```
A-SMS> show inte
interface  - Show a specific interface's configurations
interfaces - Lists all interfaces
A-SMS> show interface e
A-SMS> show interface eth0
state on
mac-addr 00:0c:29:6f:3b:e9
type ethernet
link-state link up
mtu 1500
auto-negotiation off
speed 10G
ipv6-autoconfig Not configured
monitor-mode Not configured
duplex full
link-speed 10G/full
comments
ipv4-address 10.1.1.101/24
ipv6-address Not Configured
ipv6-local-link-address Not Configured

Statistics:
TX bytes:14143922 packets:11998 errors:0 dropped:0 overruns:0 carrier:0
RX bytes:838486 packets:5962 errors:0 dropped:0 overruns:0 frame:0
A-SMS> _
```

Fig:A5 interface eth0 of A-SMS

Step 6: Check access from Windows

Command used on windows terminal

ping 10.1.1.101

```
C:\Users\Ethick_Fahad>ping 10.1.1.101

Pinging 10.1.1.101 with 32 bytes of data:
Reply from 10.1.1.101: bytes=32 time<1ms TTL=64
Reply from 10.1.1.101: bytes=32 time<1ms TTL=64
Reply from 10.1.1.101: bytes=32 time<1ms TTL=64
Reply from 10.1.1.101: bytes=32 time<1ms TTL=64

Ping statistics for 10.1.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Successful Reply from the A-SMS server

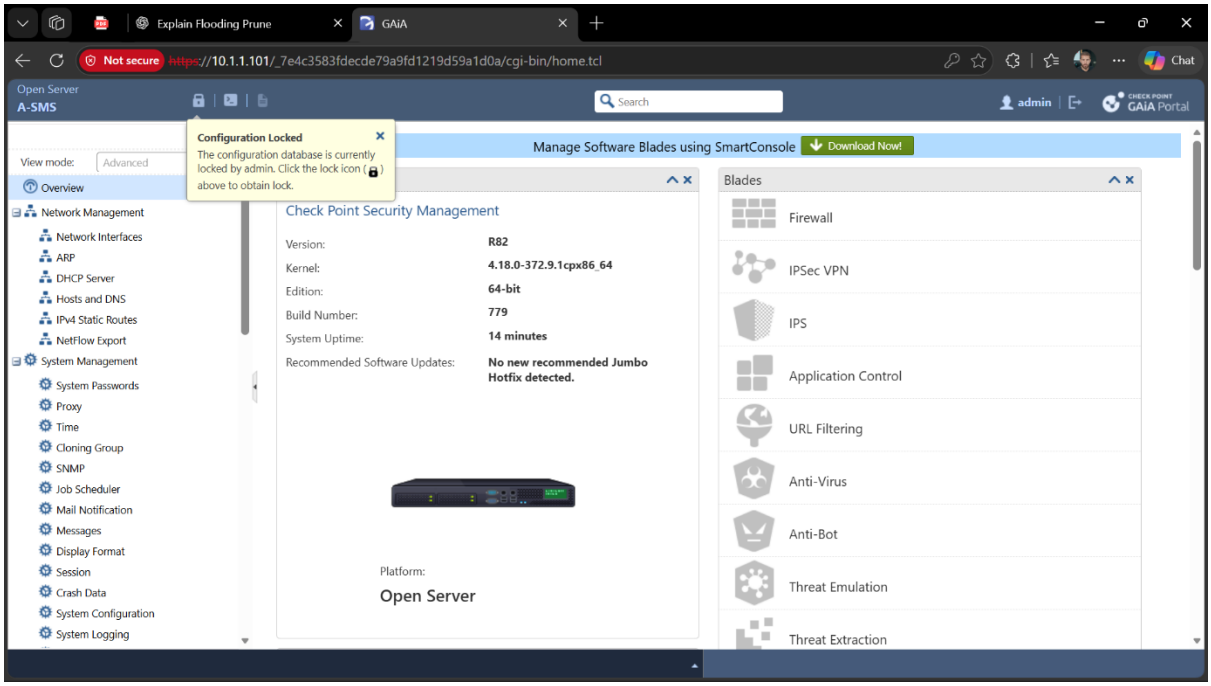


Fig A6 successful login to Gaia portal

Part A Connect SmartConsole

Step 7 Verify the server and sign in

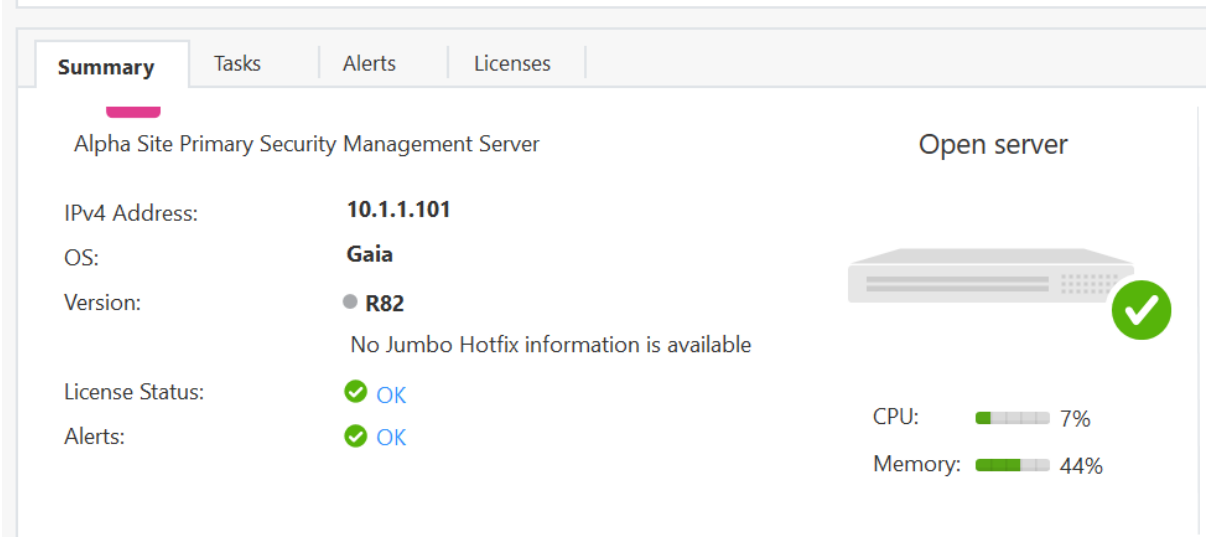
```
A-SMS> expert
Password:


Warning! All configurations should be done through clish
You are in expert mode now.

[Expert@A-SMS:0]# cp_conf finger get

HER PAR BERT NOD BUST DOWN FOOT SANG EGG SAY BASE FOLD

[Expert@A-SMS:0]# _
```



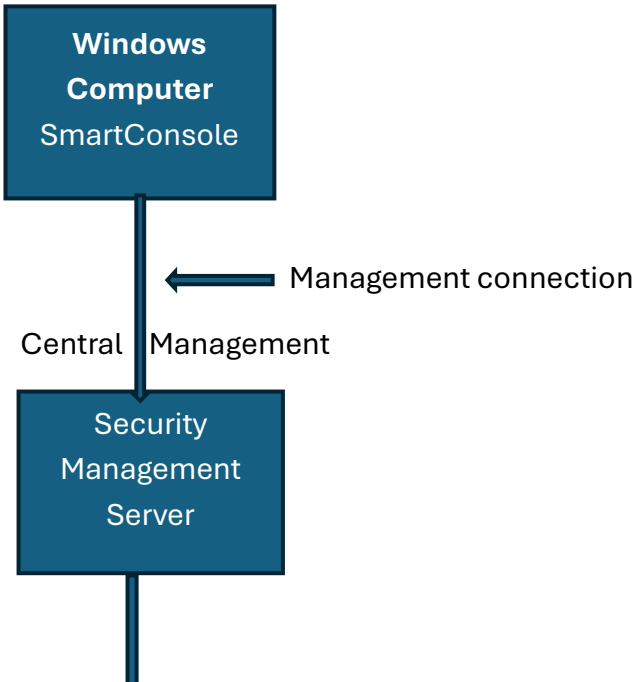
Implementation

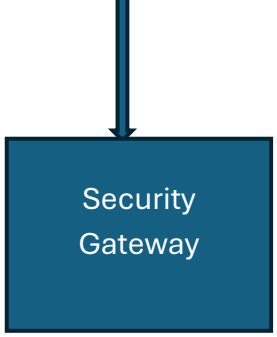
Successful authentication confirms management connectivity and server identity verification

Part B: Explain the Check Point architecture

B1: Explain the architecture

Tier 1: Management interface





Tier 3: Security Enforcement

i, SmartConsole (Management Interface):
SmartConsole is a graphical User Interface management application installed on a Windows computer. It allows network administrators to connect to the Security Management Server and manage the Check Point security environment.
Through SmartConsole, administrators can create and modify security policies, define network objects such as hosts, networks and services, monitor security activities, and manage Security Gateways.
SmartConsole acts as the interface between the administrator and the Security Management Server. It does not directly enforce security policies on network traffic.

ii. Security Management Server (SMS):
The Security Management Server is the central management component of the Check Point architecture. It maintains the security management database, including security policies, network objects, administrator information and gateway configurations.
It receives administrative changes made through SmartConsole, stores the configurations and manages the installation of security policies on the Security Gateways.
The SMS provides centralised management, making it possible for administrators to manage multiple Security Gateways from one management location.

Iii, Security Gateway:
The Security Gateway is the enforcement component of the architecture. It protects the network by inspecting incoming and outgoing traffic and applying the security policies installed from the Security Management Server.
It determines whether network traffic should be allowed or blocked based on the configured rules. It can also provide additional security functions such as Network Address Translation (NAT), VPN and threat prevention, depending on the configuration and licences.
Unlike SmartConsole and the SMS, the Security Gateway performs the actual traffic inspection and policy enforcement.

Security policies and network object definitions are maintained in the Security Management Server (SMS) database.
Administrators use SmartConsole to create and edit these configurations, but the SMS stores and manages them centrally.

Changing a policy and enforcing a policy are two different processes in Check Point architecture.

Changing a policy means creating, editing or modifying security rules and network objects through SmartConsole. The changes are saved in the Security Management Server after the administrator publishes the session.
Enforcing a policy means installing the published security policy from the SMS onto the Security Gateway. Once successfully installed, the Security Gateway applies the rules when inspecting network traffic.

B2: Explain management services and deployment

Explain three Security Management Server functions;
a, Logging and Monitoring:
The SMS collects and manages security logs generated by Security Gateways. It allows administrators to monitor network activities, review security events, identify suspicious traffic and generate reports. This helps administrators understand what is happening within the network and respond to possible security threats.

b, Security Policy Management
The SMS manages the security policies used to control network traffic. It allows administrators to create, modify and manage firewall rules that determine which traffic should be allowed or denied. It also installs these policies on Security Gateways for enforcement.

c, Network Object Management
The SMS maintains and stores network object definitions such as hosts, networks, services and Security Gateways. These objects make it easier for administrators to create and manage security rules without repeatedly entering IP addresses and other details.

Comparison Between On-Premises Management and Smart-1 Cloud.

Check Point provides different deployment options for managing its security environment. Two examples are on-premises management and Smart-1 Cloud.

On-premises management:
The management server is installed within the organization's pemises, while it provides and maintains the required hardware. Provides direct control over the management server and its environment.

Smart-1 Cloud:
Security management is provided through check point's cloud service. Check Point manages the cloud management infrastructure and reduces the organisation's responsibility for maintaining management hardware and infrastructure. Allowing administrators to manage security policies and monitor gateways through a cloud-based service.

Recommendation: For an organisation that wants to reduce on-premises hardware maintenance, Smart-1 Cloud is the recommended option.
Because Check Point provides the management service through the cloud, reducing the need for the organisation to purchase, install and maintain a dedicated management server. It also provides centralised security policy management, log analysis and reporting.

Trade-off: Smart-1 Cloud depends on reliable internet connectivity and the availability of the cloud service. The organisation also has less direct control over the underlying management infrastructure compared to an on-premises deployment.

B3 Choose the correct administration interface

Choose and justify the tool for each task.
i) inspect eth0 and its subnet mask: Gaia Clish
Explanation:Clish provides network interface information and configuration details quickly with networking commands.

ii) Review an Access Control rule: SmartConsole
Explantion: Access Control policies are managed and viewed through SmartConsole.

iii) Inspect operating-system resource use: Expert Mode
Explanation: Expert mode provides Linux commands such as top, free, and df for checking CPU, memory, and disk usage.

iv) View security logs: SmartConsole
Explanation: Logs and monitoring features in SmartConsole allow administrators to view security events and traffic logs.

Difference Between Gaia Clish, Expert Mode, and SmartConsole Command Line Panel

Gaia Clish

- Check Point's administrative command shell.
- Used mainly for network and system configuration.
- Provides structured commands and prevents many configuration errors.

Expert Mode

- Gives direct access to the underlying Linux operating system.
- Used for advanced troubleshooting, diagnostics, and resource monitoring.
- Supports standard Linux commands.

SmartConsole Command Line Panel

- A command-line interface accessed through SmartConsole.
- Allows administrators to run commands remotely on managed Check Point devices.
- Does not require a separate SSH login to the gateway.

B4: Explain security features and policy handling

1. Network Security
   A company wants to control which traffic can enter or leave its network. Network Security enforces firewall rules to allow legitimate traffic and block unauthorized access, protecting internal systems from attacks.
2. Threat Prevention
   A company wants to prevent malware, exploits, and other cyber threats from reaching users and servers. Threat Prevention inspects traffic and automatically detects and blocks malicious content before it can cause damage.

Publishing a Management Session
- Saves the administrator's changes to the Management Server.
- Makes the changes visible to other administrators.
- The changes are not yet active on the Security Gateway.

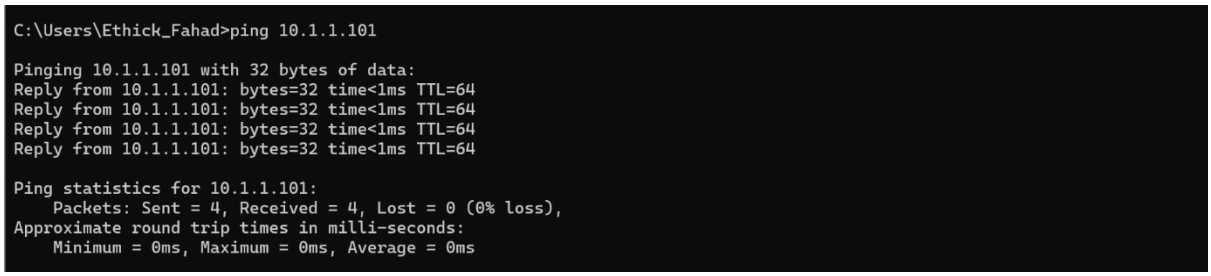
Installing Policy on a Gateway
- Sends the published security policy to the Security Gateway.
- The gateway begins enforcing the new rules.
- Until the policy is installed, the gateway continues using the previous policy.

B5. Interpret Your Own Lab Evidence
Private Networking and DHCP

The VMnet8 private network addresses can remain fixed even when the Wi-Fi network changes because VMnet8 is a virtual network created by VMware. It operates independently of the physical Wi-Fi connection, so changes to the Wi-Fi network do not affect the VMnet8 IP addresses.

Enabled VMware DHCP does not alter the static A-SMS address because the A-SMS interface is manually configured with a static IP address. DHCP only assigns addresses to devices configured to obtain an IP address automatically.

Connectivity Evidence.
A successful ping proves that basic IP connectivity exists between devices. However, a SmartConsole login proves not only network connectivity but also that the Security Management Server and required services are running and accepting management connections.



Reasoned Interpretation of Status and Logs.

An unavailable gateway object does not automatically mean the gateway has failed. The issue could be caused by network connectivity problems, incorrect configuration, service issues, or communication delays. Additional evidence is needed before diagnosing the fault.

Similarly, an empty log view does not necessarily indicate that security events are not occurring. It may result from log filters, lack of generated traffic, logging settings, or communication issues between the gateway and management server. Therefore, further investigation and evidence are required before reaching a conclusion.

Part C Explore Gaia Portal

C1: Inspect the system overview

Host Name: A-SMS
Build: 779
Uptime: 1h27min
CPU: 77%
memory: 4%

Why resource readings can change:
CPU and memory usage change continuously because the system processes different amounts of network traffic, user activity, security inspections, logging, and background services. As workload increases or decreases, resource utilization changes accordingly.

System Overview

Check Point Security Management

Version:	**R82**
Kernel:	**4.18.0-372.9.1cpx86_64**
Edition:	**64-bit**
Build Number:	**779**
System Uptime:	**1 hour 27 minutes**
Recommended Software Updates:	**No new recommended Jumbo Hotfix detected.**



Platform:

Open Server